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Module 1

Principles of Renewable Energy

The sources of energy which are being produced continuously in nature and are inexhaustible are called renewable sources of energy (or) non-conventional energy.
Some of these sources are:

- Solar energy
- Wind energy
- Bioenergy
- Ocean energy
- Geo thermal

Solar Energy

The sun is a continuous fusion reactor in which hydrogen combines to form helium and liberates large amounts of heat in the process. The sun rays contain a large amount of energy in the form of electromagnetic radiation due to the continuous nuclear fusion reaction taking place in the sun. The energy is released at the rate of 3.7 × 1020 MW. This heat energy contained in the sun rays can be utilised to generate electrical power.

Advantages

- Solar energy is available freely in nature.
- It is a renewable energy resource.
- It does not pollute the environment.
- It can be directly converted into electricity by employing photovoltaic cells.

Disadvantages

- It is available only during daytimes and clear days.
- Solar energy obtainable also depends on seasonal variations.
- It requires a large area to entrap appreciable solar energy for the generation of an economical amount of electricity.

Wind Energy

Wind is induced in atmosphere by uneven heating of earths surface by the sun. The wind energy is associated with the movement of large masses of air from cold to hot regions. The motion results from uneven heating of atmosphere by sun, thereby creating, temperature, density and pressure differences. The wind energy can be used to run windmill, which in turn will drive a generator to produce electric power or run water pumps.
The power available in the winds over the earth surface is estimated to be 1.6×10^7 MW, which is more than the present energy requirement of the world. Wind power has emerged as the most economical of all renewable energy sources. The installation cost of wind power is Rs.4 crores / MW which is comparable to that of conventional thermal plants.

Advantages

- It is freely and abundantly available in nature.
- It is a renewable energy source.

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- It does not cause pollution to environment.
- Windmills require minimal maintenance and operating cost.

Disadvantages

- It cannot produce steady and consistent power.
- It can generate only low power.

Tidal Energy

Ocean waves and tides contain a large amount of both potential and kinetic energy which can be utilised for power generation. A tide is the periodical rise and fall of sea water caused principally by the interaction of the gravitational fields of the sun and the moon. The highest level of tidal water is called flood or high tide and the lowest level is called low or ebb tide. The level difference between the high and the low tide is called tidal range. The up and down movement of the tide is used for filling and emptying the tidal basin of the plant. The tidal basin is filled up during high tide and it is emptied out during low tide. The flowing in and flowing out water between sea and tidal basin is used to run a turbine and generate electricity.

Advantages

- It is free from pollution.
- It is superior to hydel energy as it does not depend on rains.
- The tidal basin can also be used for fish farming.
- It is best suited to meet peak power demands.

Disadvantages

- Tidal power plant is costly compared to thermal and hydel power plants.
- Limited locations are available for the construction of tidal power stations.
- Power generation is not continuous and depends on the capacity of tidal basin.

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Geothermal Energy

Geothermal energy comes from the natural heat of the Earth primarily due to the decay of the naturally radioactive isotopes of uranium, thorium and potassium. Geothermal energy is derived from huge amount of stored thermal energy in the interior of earth. However, its economic recovery on the surface of the earth is not feasible everywhere. Its overall contribution in total energy requirement is negligible. However, it is a very important resource locally.

Advantages

- Carbon footprint of a geothermal power plant is low.
- Geothermal energy is a source of renewable energy.
- Does not fluctuate in the same way as other energy sources, such as solar and wind

Disadvantages

- Geothermal energy is that it is location specific. Geothermal plants need to be built in places where the energy is accessible
- Geothermal energy also runs the risk of triggering earthquakes. This is due to alterations in the Earth's structure as a result of digging.
- Geothermal energy is an expensive

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Ocean Energy

There are two types of energy that can be produced by the ocean: ***thermal energy from the sun's heat and mechanical energy from the motion of tides and waves.***
Ocean thermal energy can be converted into electricity using a few different systems that rely on warm surface water temperatures. "Ocean mechanical energy" harnesses the ebbs and flows of tides caused by the rotation of the earth and the gravitational influence of the moon. Energy from wind-driven waves can also be converted and used to help reduce one's electricity costs.
Cold ocean water from deep below the surface can be used to cool buildings (with desalinated water often produced as a by-product), and seaside communities can employ the methods to tap natural ocean energy described above to supplement municipal power and energy needs.

Advantages

- It is free from pollution.
- It is superior to hydel energy as it does not depend on rains.

Disadvantages

- High Initial Capital Investment
- Limited Locations
- The operation of commercial Tidal Power Stations has known to moderately affect the marine life around the Power Plant.
- Transportation of Tidal Energy can be expensive.

Biomass Energy

Bioenergy is a type of renewable energy derived from biomass to create heat and electricity or to produce liquid fuels such as ethanol and biodiesel used for transportation. Biomass refers to any organic matter coming from recently living plants or animals. Even though bioenergy generates about the same amount of carbon dioxide as fossil fuels, the replacement plants are grown as biomass to remove an equal amount of CO_2 from the atmosphere, keeping the environmental impact relatively neutral.

Advantages

- Bioenergy is a widely available, reliable
- Renewable source of energy.
- Harvesting biomass for electricity can also help us reduce waste.

Disadvantages

- Compared to other sources for electricity, biomass can be expensive to gather, transport, and store.

Advantages of Renewable Energy Source

- They are non-polluting
- Renewable source of energy
- Less running cost

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- They are cheap and clean
- The energy sources like wind, solar , heat waves ect. can be stored in original form
- Renewable energy sources have less capital cost
- Eco-friendly

Disadvantages of Renewable Energy

- High cost of harnessing energy
- Uncertainty of availability
- Difficulty in transporting
- Biogas energy causes greenhouse effect
- Tidal energy is difficult to harness
- Wind energy causes noise pollution
- Wind mills are costly to take-up
- Low efficiency levels
- Renewable energy can be unreliable. It changes depending upon the weather conditions

Comparison of Conventional and Non-Conventional Energy Resources

Conventional	***Non-conventional***
Traditional	Non Traditional
They can be exhausted due to over consumption.	They cannot be exhausted.
They pollute environment, on a large scale and adds to global warming.	They are environment friendly sources, which does not causes pollution.
They are primarily used for industrial and commercial purposes.	They are mainly used for domestic purposes.
These resources can be easily converted into mechanical energy	These resources require some costly methods to be converted into mechanical energy.
E.g coal, petrol, diesel etc.	E.g. Solar, wind, geothermal etc.

Energy and Sustainable Development

Global environment degradation is one of the most serious threats facing mankind as a result of the expansion of its activities around the globe. One of the international responses to global environmental problems-the framework convention on climate change- was ratified and came into effect in march 1994. The convention aims not only at stabilizing CO_2 emissions in developed countries but also at ultimately reducing man-made CO_2 emissions globally so as to stabilize the global climate. However, with fossil fuels comprising nearly 90 per cent of primary energy sources in the world, the final target of the framework convention seems very ambitious.

Environmental degradation cannot be singled out as an independent matter among various global issues. Also important are the interactions among economic development, stable energy supplies, and globe environmental conservation. In the next few decades fossil fuels will continue to be the principal source of energy driving economic development. The source of

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fossil fuels is stable and their extraction is affordable. Attempts to restrict the use of fossil fuels for environmental reasons are likely to have a negative impact on economic development and the overall availability of energy. Thus the 'three E's - environment, energy, and economic development are closely interrelated in a complex manner.

The strategy for mitigating three E's issues is a strategy for environmentally sustainable economic development. Herman Daly, a famous ecological economist, laid down three conditions for sustainability;

1. The consumption rate of renewable resources is not higher than its recovery rate.

2. The consumption rate of non-renewable resources is not higher than the rate of increase in renewable resource supply.

3. The emission of pollutants is within the absorption capacity of the environment.

Unfortunately, these conditions have been violated for years. Examples of respective violations typically include deforestation, the depletion of fossil fuels, and the increase in CO_2 concentration in the air. Such violations may be hard to reverse in the short term but, unless long-term remedial action is taken, present global development trends will not be sustainable. In particular, a substantial reduction in resource consumption and emissions of pollutants is essential for the development of a sustainable human society on this planet. As evident from the above discussions, economy, environment and energy are closely interrelated and an overall policy is required to deal with them.

UNFCC

UNFCC stands for United Nations framework convention on climate change. It was adopted on 5 June 1992.the objective of UNFCC is to stabilize greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system. It also states that such a level should be achieved with in a time frame sufficient to allow ecosystems to adopt naturally to climate change, to ensure that food production is not threatened and enable economic development to proceed in a sustainable manner.

Kyoto protocol

The Kyoto protocol is a legally binding agreement on international level accepted by most of the countries to reduce greenhouse gas emission and risk of global warming. It came in to force on 16 February 2005. About 183 countries including India have ratified this protocol.
The major feature of the Kyoto protocol is that it sets binding targets for industrialised countries to reduce greenhouse gases emission so that global warming can be monitored and controlled. The protocol recognizes and accepts that the developed countries are principally responsible for the current high concentration of greenhouse gases in the outer atmospheric layer of the earth as a result of more than 150 years of in industrial activities. Hence Kyoto protocol places a heavier burden on the developed countries. The Kyoto protocol lays down that the developed countries are required to reduce their emission of greenhouse gases by an average of 5.2% below the year 1990 levels by the year 2012.

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- ***Emission Trading or Carbon Trading***

The member countries of the protocol have accepted emission targets for limiting or reducing the greenhouse gas emissions. These emission targets are called allowed emission or assigned amounts. A country cannot generate greenhouse gases more than the assigned amount units.
The countries are allowed to use the option of emission trading or carbon trading to meet their obligations if they maintain or increases their greenhouse gas emission. Emission or carbon trading allows nations that can easily meet their targets to sell emission credits to those that cannot meet their emission targets.

The Kyoto protocol sets specific emission targets to each industrialised nation, bit it excluded developing countries from any emission target. To meet their target, member countries have to adopt the following measures.

- Place restrictions on industry causing major pollution problems.
- Manage transportation to reduce emission from automobiles
- Adopt renewable energy sources in place of fossil fuels.

- ***Clean development mechanism***

This mechanism allows a country with the emission reduction or limitation commitment to implement an emission reduction project in developing countries. Such projects can earn saleable certified emission reduction (CER) credits.
Each CER credit is equivalent to the reduction of 1 ton of CO_2 emission and this can be counted towards meeting assigned GHG emission targets. A clean development mechanism activity may be a rural electrification project using solar panel or adoption of energy efficient boilers there by reducing greenhouse gas emission.

- ***Joint Implementation***

It allows a country with an emission reduction or limitation commitments under the protocol to earn emission reduction units(ERUs) by executing an emission reduction or removal project in another country. Each ERU is equivalent to the reduction of 1 ton of CO_2 gas. The ERUs are counted towards the emission limitation target of a country.

World Wide Renewable Energy Availability

Conventional Sources

Fossil Fuels: Fossil fuels include coal, oil and gas. These fuels are major source of energy since about 1850, the start of industrial era. Fossil fuels are so called because these are in fact the fossils of old biological life that once existed on the surface of earth. It is formed in several parts of earth at varying depths, during several million years by slow decomposition and chemical actions of buried organic matter under favorable pressure, heat and bacterial marine environment.
***Coal*:** Despite its poor environmental credentials, coal remains a crucial contributor to energy supply in many countries. Coal's dominant position in the global energy mix is largely due to the fact that it is abundant, widely distributed across the globe and affordable. Coal is the

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most widespread fossil fuel around the world, found in about 75 countries and actively mined in 50 countries.

Oil: Oil is a premier energy source with a wide range of possible applications. Its main use however, will be shifting towards transport and the petrochemical sector.

Gas: The cleanest of all fossil-based fuels, natural gas is plentiful and flexible. It is increasingly used in the most efficient power generation technologies, such as, Combined Cycle Gas Turbine (CCGT) with conversion efficiencies of about 60 percent. The reserves of conventional natural gas have grown by 36 per cent over the past two decades and its production by 61 per cent. The exploration, development and transport of gas usually require significant upfront investment. Shale gas refers to natural gas that is trapped within shale formations. Shales are fine-grained sedimentary rocks that can be rich resources of petroleum and natural gas. "Shale Gas Revolution" changed the natural gas scene, first in the United States and subsequently in other countries around the globe.

Hydro Resources: Among all renewables, hydropower is the most advanced and flexible source of power. It is a well-developed and established source of electric power. Due to requirement of huge capital investment and strong environmental concerns about large plants, only about one third of the realistic potential has been taped so far. Hydro installations and plants are long lasting (turbine life is about 50 years). This is due to continuous steady operation without high temperature or other stresses. Therefore, it often produces electricity at low cost with consequent economic benefits.

Nuclear Resources: U235, U233 (isotopes of uranium) and Pu239 (plutonium) are used as nuclear fuels in nuclear reactors (thermal reactors) and are known as fissile (or fissionable) materials. Out of these, only U235 occurs in nature. Uranium reserves in the world are small and its recovery is expensive. Concentrated deposits of Uranium are not available. The content of natural Uranium in Uranium ore is about 0.1–0.5 per cent. Major available sources of Uranium are in Australia, Canada, and Kazakhstan and to a lesser extent the USA.

Non-conventional Sources

Non-conventional technologies are presently under development stage. At present its share is very small.

Solar Energy: Solar energy can be major source of power and can be harnessed by using thermal and photovoltaic conversion systems. Maximum solar radiation received on the surface of earth, normal to it, on bright sunny day, at noon is approximately 1 kW/m^2 at sea level. The earth continuously intercepts solar power of 178 billion MW, which is about 10,000 times the world's demand. Solar PV power is considered as an expensive source of power. However, the prices are decreasing steadily.

Wind Energy: The power available in the winds over the earth surface is estimated to be 1.6×10^7 MW, which is more than the present energy requirement of the world. Wind power has emerged as the most economical of all renewable energy sources. The installation cost of wind power is 4 crores / MW. There has been remarkable growth of wind power installation in the world. Wind power generation is the fastest growing energy source. China, with about 114.6 GW, has the highest installed capacity while Denmark, with over 4.8 GW, has the highest level per capita. Wind accounts for about 39 per cent of Denmark's electricity production.

Biomass Energy: Energy resources available from animal and vegetation are called biomass energy resources. Historically, humans have harnessed biomass- derived energy since the time when people began burning wood to make fire. Even today, biomass is the only source of fuel for domestic use in many developing countries. This is an important resource for developing countries, especially in rural areas. The principal biomass resources are: Trees

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(wood, leaves and forest industry waste), Cultivated plants grown for energy, Algae and other vegetation from ocean and lake, Urban waste (municipal and industrial waste) Rural waste (agricultural and animal waste, crop residue, etc.)

Geothermal Energy: Geothermal energy comes from the natural heat of the Earth primarily due to the decay of the naturally radioactive isotopes of uranium, thorium and potassium. Geothermal energy is derived from huge amount of stored thermal energy in the interior of earth. However, its economic recovery on the surface of the earth is not feasible everywhere. Its overall contribution in total energy requirement is negligible. However, it is a very important resource locally. Of the countries utilising their geothermal resource, almost all use it directly but only 24 use it for electricity generation.

Ocean Tidal Energy: Tidal energy is a form of hydropower that converts energy of tides into electricity or other useful forms of power. Although not yet widely used, tidal power has potential for future electricity generation. Tides are more predictable than wind energy and solar power. It is in developing stage. There are at present only few operational tidal power plants. The first and the biggest 240 MW tidal plant was built in 1966 in France at the mouth of La Rance river, near St. Malo on the Brittany coast.

Ocean Thermal Energy Conversion OTEC technology is in infant stage. Conceptual designs of small OTEC plants have been finalized. Their commercial prospects are quite uncertain. The potential is likely to be more than that of tidal or wave energy. Up to 88,000 TWh/yr of power could be generated from OTEC without affecting the ocean's thermal structure. While OTEC technology has been demonstrated in the ocean for many years, the engineering and commercial challenges have constrained the development of the technology.

Renewable Energy Availability in India

Overall Production and Consumption

India is both a major energy producer and consumer. India currently ranks as the world's 5th largest energy producer after China, USA, Russian Federation and Saudi Arabia, accounting for about 4 per cent of the world's total annual energy production. Also it is the world's 4th largest energy consumer trailing China, USA and Russian Federation and accounting for 5.9 per cent of the world's total annual energy consumption. Thus India is a net energy importer, mostly due to large imbalance between oil production and consumption.

Conventional

Fossil Fuel: India has vast reserves of coal, 5th largest in the world after the USA, Russia, China and Australia. According to a rough estimate, total recoverable coal in India is 60,600 million tons, about 6.8 per cent of world's total. Our coal production is not enough to meet the growing consumption. Net coal import dependency has risen from a negligible percentage in 1990 to nearly 23 per cent in 2014. Indian coal has high ash content (25–50 per cent), low heat value (3000–4000 kcal/kg) and low sulfur content (1 per cent). We have very little oil and gas reserves. The estimated reserve of crude oil is 762 MT, which is 0.34 per cent of global oil reserves. Likewise, the estimated reserve of Natural gas is 1427 bcm, which is 0.68 per cent of global gas reserves. Oil and gas reserves are insufficient even for transportation sector.

Hydro Resources: India stands 7th in the list of nations with hydro resources. Total potential is 100,000 MW. Approximately 41,997 MW has been developed (as on July 31, 2015). Huge installation cost, environmental and social problems are major difficulties in its development. Also, 56 sites for pumped storage schemes with an aggregate installed capacity of 94,000 MW have been identified.

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Nuclear Resources: India has modest (1 to 2 per cent of global) reserves of Uranium, mostly located at Jadugoda, Jharkhand. There are 21 nuclear reactors at 7 plants located at Narora, Rawatbhatta, Kakrapar, Kaiga, Kalpakkam, Kudankulam and Trombay (18 PHWR, 2BWR and 1 VVER). Present, nuclear power installed capacity is 5,780 MW, which is 2.1 per cent of total installed electricity generating capacity. (as on July 31, 2015). There are six reactors totaling 4,300 MW under construction in Tamil Nadu, Gujarat and Rajasthan.

Non-conventional

Located in the tropical region, India is endowed with abundant renewable energy, i.e., solar, wind and biomass including agricultural residue which are perennial in nature. Harnessing these resources is best suited to meet the energy requirement in rural areas in a decentralized manner. India has potential of generating more than 900 GW (assuming 3 per cent wasteland is made available) from commercially exploitable renewable energy resources.

Wind Energy: The wind power program in India was initiated in 1983–84. The highly successful India's wind power programme is entirely market driven. This sector has been growing very fast for the last few years. India currently (year 2015) stands 5th in the world among countries having large installed capacity wind generators, after China, USA, Germany and Spain. Current (July 31, 2015) installed capacity for wind power stands at 23,864.9 MW, mostly located in Tamil Nadu, Gujrat, Maharashtra and Rajasthan.

Solar Energy: India, receives solar energy equivalent to more than 5,000 trillion kWh per year, which is far more than its total annual consumption. The average daily global radiation is around 5 kWh per sq. m per day with the sunshine ranging between 2300 and 3200 hours per year in most parts of India.

Biomass Energy: A large quantity of biomass is available in our country in the form of dry waste like agro residues, fuel wood, twigs etc. and wet wastes like cattle dung, organic effluents, sugarcane bagasse, banana stem etc. Potential for generation of electric power/cogeneration is 25 GW from agro residue and bagasse. The potential from urban waste is 1,700 MW. Also there is vast scope for production of bio diesel.

Small Hydro Resources: Hydro resources of capacity less than 25 MW are called small, less than 1 MW are called mini and less than 100 kW are called micro hydro resources. Total potential is 20 GW out of which 4,130 MW has been realized.

Geothermal Energy: Potential in geothermal resource in the country is 10,600 MW of electric power generation. As a result of various resource assessment studies/surveys, nearly 350 potential hot springs have been identified throughout the country. Most of them are low temperature hot water resources and can best be utilized for direct thermal applications. Only some of them can be considered suitable for electrical power generation.

Ocean tidal Energy: There is no functional tidal plant at present. The identified economic tidal power potential has been estimated as 8,000–9,000 MW. Three sites have been identified for development of tidal energy.

- Gulf of Kutch: potential = 1200 MW
- Gulf of Cambay (Khambat): Potential = 7,000 MW
- Sundarbans: potential = 100 MW

Emerging Technologies Emerging technologies like 'fuel cell' and 'hydrogen energy' are suited for stationary and portable power generation, which suits for transportation purposes. In view of the growing importance of Fuel cells and Hydrogen, a National Hydrogen Energy Board has been created.

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Wind Energy

Introduction

Wind energy is the kinetic energy associated with movement of large masses of air.

These motions result from uneven heating of atmosphere by the sun, creating temperature, density and pressure differences. It is estimated that 1 percent of all solar radiation falling on the face of the earth is converted into kinetic energy of the atmosphere, 30 per cent of which occur in the lowest 1000 m of elevation. It is thus an indirect form of solar energy.

Wind energy is harnessed as mechanical energy with the help of wind turbine. The mechanical energy thus obtained can either be used as such to operate farm appliances, water pumping, etc., or converted to electric power and used locally or fed to a grid. A generator coupled to wind turbine is known as aero-generator.

Advantages

- wind energy can be available continuously throughout 24-hour day, though it can vary to a great extent including no-wind periods.
- It is a clean, cheap, and eco-friendly renewable source.

Disadvantages

- It is a dispersed, erratic and location-specific source.
- Very slow winds are useless, having no possibility of power generation.

On the other hand, very strong stormy winds cannot be utilized due to safety of turbine. Moderate to high-speed winds, typically from 5 m/s to about 25 m/s are considered favorable for most wind turbines.

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Properties of wind

- Power available in wind increases with the cube of its speed, meaning even a small rise in speed greatly increases energy potential.
- Denser air, which occurs in colder or high-pressure regions, carries more energy.
- Wind formation depends on the temperature gradient, where differences in temperature create pressure variations that drive air movement.
- Winds always flow from high-pressure to low-pressure areas, and a larger pressure difference produces stronger winds.
- Turbulence is a property that describes irregular and unstable airflow caused by terrain or obstacles, which can reduce turbine efficiency.
- Altitude influences wind behavior, as wind speeds generally increase with height due to reduced friction near the ground.
- Finally, wind has seasonal variations, with its availability changing according to monsoon patterns and climatic conditions.

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Availability of wind energy in India

Wind energy in India is emerging as an important and reliable renewable energy source. Wind energy is generated by wind turbines installed in wind farms. According to the Ministry of New and Renewable Energy (MNRE), the total installed capacity of Wind Energy in India is 41.2 GW (as of September 2022). There are more than 800 wind power plants in India installed to generate wind power.

India's installed wind power capacity is the 4th largest in the world. The wind power capacity is mainly spread across, southern, western and north-western regions of India. Wind Energy holds the major portion of 34.06% of total RE capacity among renewable and continues as the major supplier of clean energy.

70% of annual wind generation is during the five-month duration from May to September coinciding with Southwest monsoon duration. The government of India has fixed a target of 500 GW of Renewable Energy by 2030 out of which 140 GW will be from Wind.

Overall the future of Wind Energy in India is bright as energy security and self-sufficiency is identified as the major driver. Wind power accounts for nearly 10% of India's total installed utility power generation capacity and generated 71.814 TWh in the year 2022–23, which is nearly 4.43% of total electricity generation.

The Indian government has installed over 800 wind-monitoring stations all over the country through the National Institute of Wind Energy (NIWE) and issued wind potential maps at 50m, 80m, 100m, and 120m above ground level.

Gujarat & Tamil Nadu are the two major wind power generating states in India.

The Gujarat government's focus on tapping renewable energy has led to a sharp rise in wind power capacity in the last few years. Gujarat has the highest share (around 24%) of the total installed wind power capacity of the country.

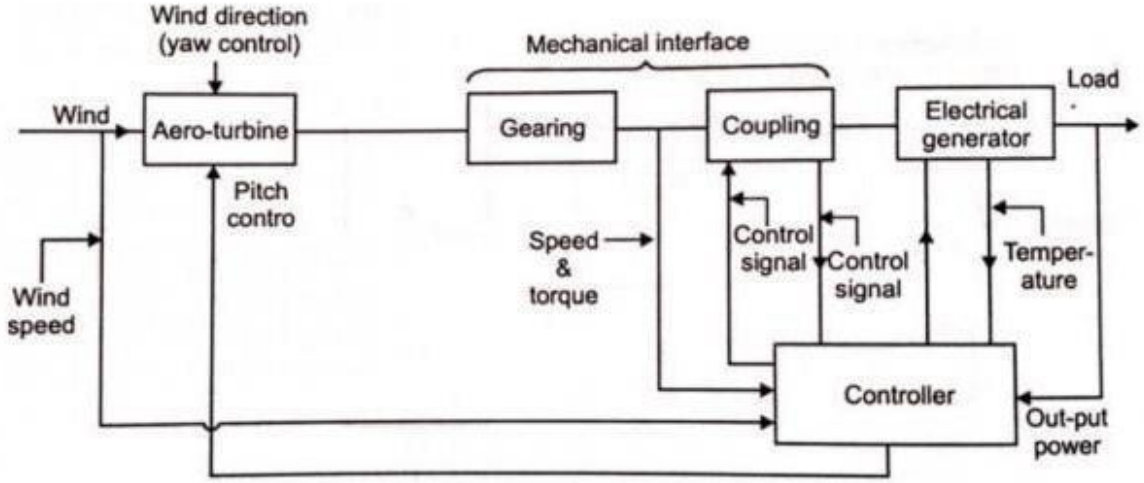
Tamil Nadu's wind production capacity was 2nd highest (around 23% share) after Gujarat as of 31 May 2023.

Major factors that have led to accelerated development of the wind power are as follows:

- Availability of high-strength fiber composites for constructing large low-cost rotor blades
- Falling prices of power electronics
- Variable speed operation of electrical generators to capture maximum energy
- Improved plant operation, pushing the availability up to 95 percent
- Economy of scale, as the turbines and plants are getting larger in size
- Accumulated field experience (the learning curve effect) improving the capacity factor
- Short energy payback (or energy recovery) period of about one year

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Wind Energy Conversion System



A wind energy conversion system converts wind energy into some form of electrical energy.

In particular, medium and large scale WECS are designed to operate in parallel with a public

or local ac grid. This is known as grid-connected system. A small system, isolated from grid, feeding only to local load is known as autonomous, remote, decentralized, stand alone or isolated power system

Wind energy extracted by the blades, rotates the shaft, which by using the gear and coupling mechanism, operates the generator housed inside the nacelle.

Aero turbines convert energy in moving air to rotary mechanical energy. They require pitch and yaw control. ***Yaw control***: as wind direction changes a motor rotates turbine slowly so as to face the blades into the wind. ***Pitch control***: pitch angle is the incident angle of wind with blade. Max power when pitch angle is zero. This control provided by pitch control. A mechanical interface consisting of a step up gear and a suitable coupling transmits the rotary mechanical energy to an electric generator. Output of this generator connected to load or power grid. The purpose of controller is to sense various factors as necessary and appropriate Control signals for matching electrical output to wind energy input and protect the System from extreme conditions brought by strong winds, electrical faults etc.

The main components are,

- Aeroturbine
- A mechanical interface
- generator
- controller

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Aeroturbine

It converts energy in moving air to rotary mechanical energy. In general, they require pitch control (When the power output becomes too high, it sends an order to the blade pitch mechanism which immediately pitches (turns) the rotor blades slightly out of the wind. Conversely, the blades are turned back into the wind whenever the wind drops again.) and yaw control (As the direction of the wind changes frequently, the yaw control is provided to steer the axis of the turbine in the direction of the wind) for proper operation.

Mechanical interface

Consisting of a step up gear and a suitable coupling transmits the rotary mechanical energy into electrical generator. This block acts as a drive for the generator. Use of variable gear ratio has been considered in the past and was found to add more problem than benefit.

Generator

Large WTG used with grid connected system have induction generators. Medium capacity uses synchronous generators. Small capacity uses PMMDC generators. The output of this generator is connected to the power /load as the application warrants.

Controller

The purpose of the controller is to sense wind speed, wind direction, shaft speed and torque at one or more points, output power, and generator temperature as necessary and protect the system from extreme conditions brought up by strong winds, electrical faults.

The functions are,

- The orientation of the rotor to the wind
- Start up and cut in of the equipment
- Power control of the rotor by varying the pitch of the blades
- Generator output monitoring status, data computation and storage
- Shut down and cut out owing to malfunction of every high wind

The main components of control system are,

- Sensor - mechanical, electrical or pneumatic
- Decision elements - relays, logic modules, analogue circuits, microprocessors, mechanical units etc.
- Actuators - hydraulic, electric or pneumatic

Towers

Four type of towers are used

- Reinforced concreate towers
- The pole tower
- The truss tower

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- The built up shell tube tower

Based on the generator drive, two schemes have been developed for the operation of WECS:

- Fixed Speed Drive Scheme and
- Variable Speed Drive Scheme.

Applications Requiring Mechanical Power

***Wind pumps*:** Low power turbines are used for producing mechanical power for pumping water in remote areas. Simple and reliable traditional reciprocating pumps or centrifugal pumps are used. These wind pumps are used to supply water for livestock, small-scale irrigation, low head pumping for aquatic breeding and domestic water supply.

Heating: The direct dissipation of mechanical power produces heat with 100 per cent efficiency using paddlewheel and other turbulent fluid systems. The available hot water is used as such or employed for space heating.

***Sea transport*:** The old square-rigged sailing ships were inefficient as they were operated by drag forces (to be discussed later). Modern racing yachts, with a subsurface keel, harness lift forces and are much more efficient and can sail faster than the wind. Large cargo ships requiring power in MW range, driven by improved efficient sails are now being designed. Also wind turbines are installed onboard to power propellers in ferries operating on short routes.

As Off Grid Electrical Power Source

Machines of low power with a rotor diameter of about 3 m and 40–1000 W rating can generate sufficient electrical energy for ***space heating and cooling of homes, water heating, battery charging, and for operating domestic appliances such as fan, light and small tools.***

Applications for somewhat more powerful turbines of about 50 kW are: ***producing electrical power for navigation signal (e.g. lighthouse), remote communication, weather stations and offshore oil drilling platforms.***

Intermediate power range, roughly 100 to 250 kW aero-generators can ***supply power to isolated populations, farm cooperatives, commercial refrigeration, desalination and to other small industries.*** The generator may operate in stand-alone mode or may be connected to a mini-grid system.

For lifting water to a hill, aero-generator is installed on the top of hill and electrical energy is transmitted to a pump fixed at a lower level. Same principle is utilized to ***store excess generated power using pumped storage system, to be utilized later during no wind periods***.

As Grid Connected Electrical Power Source

Large aero-generators in the range of few hundred kW to few MW are planned for ***supplying power to a utility grid***. Large arrays of aero-generators, known as wind farms are being deployed in open plains or offshore in shallow water for this purpose

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Wind Turbine Types And Their Construction

Wind turbines are broadly classified into two categories.

- Horizontal Axis Wind Turbine (HAWT)
- Vertical Axis Wind Turbine (VAWT)

When the axis of rotation is parallel to the air stream (i.e. horizontal), the turbine is said to be a Horizontal Axis Wind Turbine (HAWT), and when it is perpendicular to the air stream (i.e. vertical), it is said to be a Vertical Axis Wind Turbine (VAWT). The size of the rotor and its speed depends on the rating of the turbine.

Horizontal Axis Wind Turbine (HAWT)

HAWTs have emerged as the most successful type of turbines. These are being used for commercial energy generation in many parts of the world. Their theoretical basis is well researched and sufficient field experience is available with them.

Main Components

The constructional details of most common, three-blade rotor, horizontal axis wind turbine are shown in Fig. Main parts are:

- Turbine Blades
- Hub
- Nacelle
- Yaw Control Mechanism
- Tower

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Turbine Blades: Turbine blades are made of high-density wood or glass fiber and epoxy composites. They have air foil type cross-section. The blades are slightly twisted from the outer tip to the root to reduce the tendency to stall.

In addition to centrifugal force and fatigue due to continuous vibrations there are many extraneous forces arising from wind turbulence, gust, gravitational forces and directional changes in the wind, etc. All these factors are to be taken care off at the designing stage. Diameter of a typical, MW range, modern rotor may be of the order of 100 m. Modern wind turbines have two or three blades. Two/three blade rotor HAWT are also known as propeller type wind turbines owing to their similarity with propellers of old aero planes. However, the rotor rpm in case of wind turbine is very low as compared to that for propellers.

The relative merits and demerits of two and three blade rotors are as follows:

- Compared to two-blade design, the three-blade machine has smoother power output
- It also has balanced gyroscopic force.
- There is no need to teeter (to be discussed later in this section) the rotor, allowing the use of simple rigid hub.

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- The blades may be cross-linked for greater rigidity.
- Adding third blade increases the power output by about 5 per cent only, while the weight and cost of rotor increases by 50 per cent, thus giving a diminished rate of return for additional 50 per cent weight and cost.
- The two-blade rotor is also simpler to erect, since it can be assembled on ground and lifted to the shaft without complicated maneuvers during the lift.

Hub: The central solid portion of the rotor wheel is known as hub. All blades are attached to the hub. Mechanism for pitch angle control is also provided inside the hub.

Nacelle: The term nacelle is derived from the name for housing containing the engines of an aircraft. The rotor is attached to nacelle, mounted at the top of a tower. It contains rotor brakes, gearbox, generator and electrical switchgear and control. Brakes are used to stop the rotor when power generation is not desired. Gearbox steps up the shaft rpm to suit the generator. Protection and control functions are provided by switchgear and control block. The generated electrical power is conducted to ground terminals through a cable.

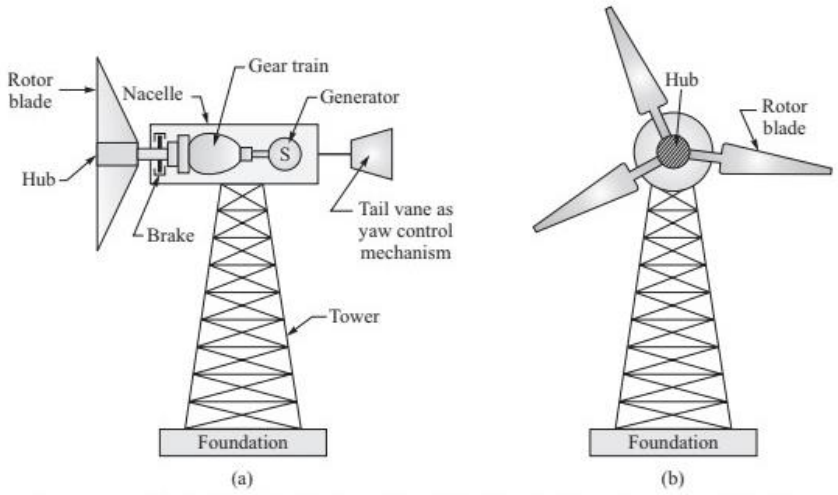
Yaw Control Mechanism: The mechanism to adjust the nacelle around vertical axis to keep it facing the wind is provided at the base of nacelle.

Tower: Tower supports nacelle and rotor. For medium and large sized turbines, the tower is slightly taller than the rotor diameter. In case of small sized turbine, the tower is much larger than the rotor diameter as the air is erratic at lower heights. Both steel and concrete towers are being used. The construction can be either tubular or lattice type. The tower vibrations and resulting fatigue cycles under wind speed fluctuations are avoided by careful design. This requires avoidance of all resonance frequencies of tower, the rotor and the nacelle from the wind fluctuation frequencies.

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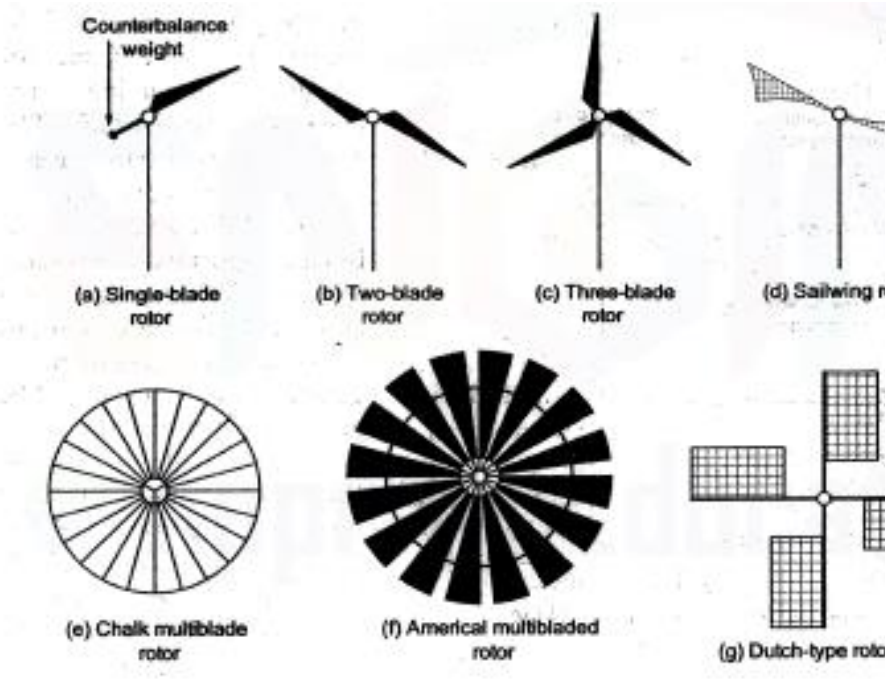
Types of Rotors

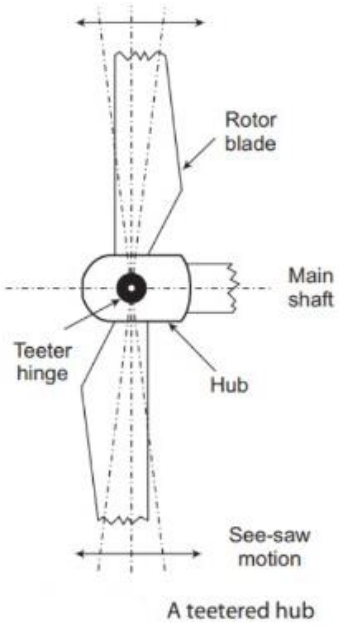
Depending on the number of blades, wind speed and nature of applications, rotors has been developed in various types of shapes and sizes.



Wind turbine (a) Side view of the wind turbine, (b) Front view of the wind turbine.

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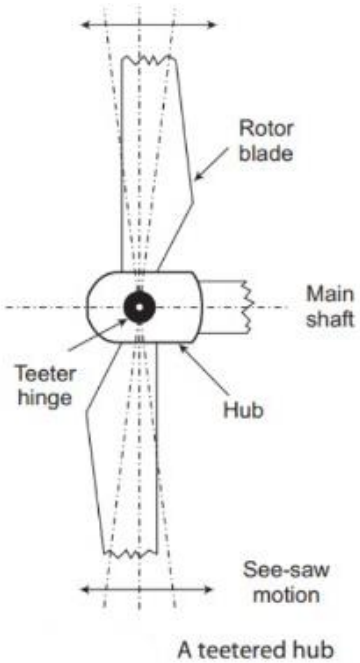
The types of rotors shown in (a) to (e) are relatively high-speed ones, suitable for applications such as electrical power generation. Large HAWTs have been manufactured with two and three blades.

A single-blade rotor, with a balancing counterweight is economical, has simple controls but it is noisier and produces unbalanced forces. It is used for low-power applications.
Those given in Fig. (f) and (g) are low-speed rotors and most suited for water lifting applications, which require highstarting torque. They can capture power even from very slow winds.

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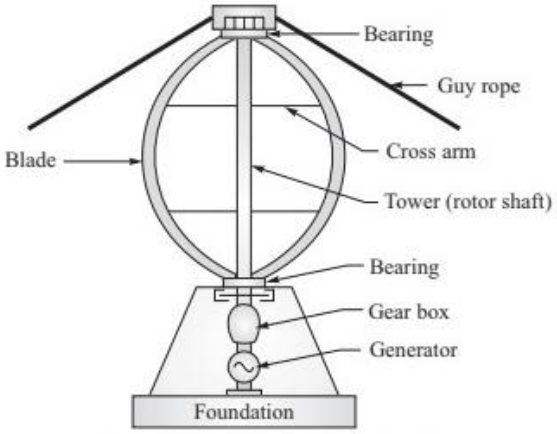
Teetering of Rotor

As wind speed rises with height, the axial force on blade when it attains the upper position is significantly higher as compared to that when it is at lower position. For one and two blade rotors this causes cyclic(sinusoidal) load on a rigid hub leading to fatigue



This is greatly relieved by providing a teeter hinge (a pivot within the hub) that allows a see-saw motion to take place out of the plane of rotation (i.e. vertical plane). The rotor leans backwards to accommodate the extra force as shown in Fig. This also reduces blade loads near the root by approximately 40 per cent. The use of third blade has approximately same effect as a teeter hinge on the hub moments since the polar symmetry of the rotor averages out the applied sinusoidal loads. Therefore, teetering is not required when the number of blades is three or more.

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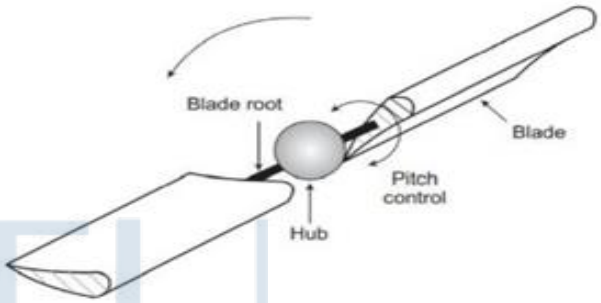


Vertical axis wind turbine.

Yaw Control System

Adjusting the nacelle about vertical axis to bring the rotor facing the wind is known as yaw control. The yaw control system continuously orients the rotor in the direction of wind. For localities with prevailing wind in one direction only, the rotor can be in a fixed orientation. Such a machine is said to be yaw fixed. Most wind turbines however, are yaw active. In small wind turbines, a tail vane is used for passive yaw control. In large turbines however, anactive yaw control with power steering and wind direction sensor is used to maintain the orientation.

Pitch Control System



Pitch of a blade is controlled by rotating it from its root, where it is connected to the hub as shown in Fig. Pitch control mechanism is provided through the hub using hydraulic jack in the nacelle. The control system continuously adjusts the pitch to obtain optimal performance. In modern machines, pitch control is incorporated by controlling only outer 20 per cent length of the blade (i.e. tip), keeping remaining part of the blade as fixed.

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Vertical Axis Wind Turbine (VAWT)

VAWTs are in the development stage and many models are undergoing field trial. Main attractions of a VAWT are:

- It can accept wind from any direction, eliminating the need of yaw control.
- Gearbox, generator etc. are located at the ground, thus eliminating the heavy nacelle at the top of the tower.
- This simplifies the design and installation of the whole structure, including tower.
- The inspection and maintenance also gets easier and it also reduces the overall cost.

Main Components

The constructional details of a vertical axis wind turbine (Darrieus type rotor) are shown in Fig. The details of main components are as follows:

- Tower (or Rotor Shaft)

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- Blades
- Support Structure

Tower (or Rotor Shaft): The tower is a hollow vertical rotor shaft, which rotates freely about vertical axis between top and bottom bearings. It is installed above a support structure. In the absence of any load at the top, a very strong tower is not required, which greatly simplifies its design. The upper part of the tower is supported by guy ropes. The height of the tower of a large turbine is around 100 m.

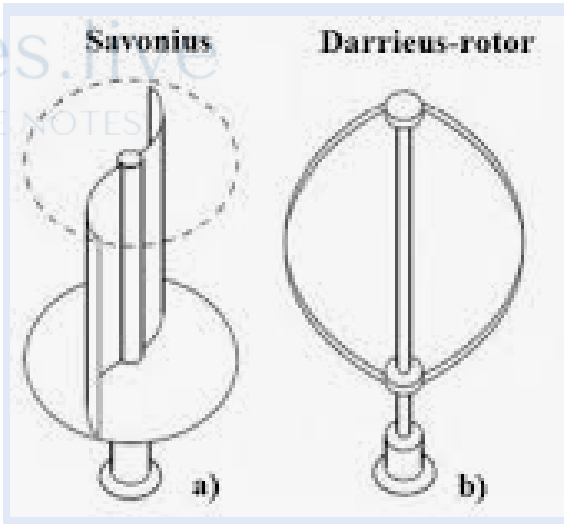
Blades: It has two or three thin, curved blades shaped like an eggbeater in profile, with blades curved in a form that minimizes the bending stress caused by centrifugal forces-the so-called 'Troposkien' profile. The blades have air foil cross-section with constant chord length. The pitch of the blades cannot be changed. The diameter of the rotor is slightly less than the tower height. The first large (3.8 MW), Darrieus type, Canadian machine has rotor height as 94 m and diameter as 65 m with a chord of 2.4 m

Support Structure: Support structure is provided at the ground to support the weight of the rotor. Gearbox, generator, brakes, electrical switchgear and controls are housed within this structure.

<u>Types of Rotors</u>

Various types or rotors for VAWTs are shown in Fig.

The ***Savonius or S-rotor*** consists of two half cylinders attached to a vertical axis and facing in opposite directions to form a two-vaned rotor. It has high starting torque, low speed and low efficiency. It can extract power even from very slow wind, making it working most of the time. These are used for low power applications. High starting torque particularly makes it suitable for pumping applications, using positive displacement pumps.



Darrieus rotor is used for large-scale power generation. Power coefficient is considerably better than S-rotor. It runs at a large tip-speed ratio. The aerodynamic force on the blade reverses in every revolution causing fatigue. This along with centrifugal force complicates the design of the blade. One of the drawbacks of this rotor is that it is usually not self-starting. Movement may be initiated by using electrical generator as motor. As the pitch of the blade cannot change, the rotor frequency and thus the output power cannot be controlled. Rotor frequency increases with wind speed and power output keeps on increasing till the blades stall. Hence at high wind speed it becomes difficult to control the output. For better performance and

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safety of the blades, gearbox and generator, etc., it is desirable to limit the output to a level much below its maximum possible value.

Savonius Rotor

Advantages

- It has vertical rotating rotor shaft which eliminates the need of any expensive transmission system from the rotor to generator.
- It produces power effectively in slow wind speeds (as low as 8 km/h).
- The generator can be located at ground level which helps in easy maintenance.
- The tower is simple in construction and installation due to lower loads.
- The cost of construction and installation is low.
- Yaw and pitch control mechanisms are not required.

Disadvantages

- It utilises drag force which results in its lower efficiency.
- Power output is very low.
- The power output is low as the pitch control of the blade is impossible.
- It is unsuitable where tall installation is necessary due to wind conditions.
- It has low power to weight ratio.

Darrieus Rotor

Advantages

- The rotor blade can accept the wind from any direction.
- It does not require any yaw control mechanism.
- The blades have constant pitch. There is no need of pitch control mechanism.
- The gear train and generator are mounted at the ground level.
- The tower is simple in construction.
- The cost of construction and installation is low.
- It has higher power coefficient and tip speed ratio compared to S-rotor.

Disadvantages

- It is not self-starting machine.
- It works on drag force and its efficiency to convert wind energy is low compared to horizontal axis rotors.
- It can have limited height. It cannot utilize high wind speeds available at higher level.
- It cannot be yawed out of the wind; special high torque braking system is required during the occurrence of high wind speeds.

Darrieus wind turbine operates on drag force created on it by wind stream. It also has full directional symmetry to capture wind energy, thereby resulting in presence of zero.

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Horizontal Axis versus Vertical Axis Turbines

Most wind turbines, used at present are of horizontal axis type. They have been well researched and have gone through extensive field trial. As a result, well-established technology is available for HAWTs.

Some advantages of VAWT have recently generated considerable interest in this type of turbine. These are:

- it can accept wind from any direction without adjustment, which avoids the cost and complexity of yaw orientation system, gearing and generators, etc., are located at ground level, which simplifies the design of tower, the installation and subsequent inspection and maintenance and
- also they are less costly as compared to HAWTs. T

The principal disadvantages of VAWTs are:

- Many vertical axis machines have suffered from fatigue arising from numerous natural resonances in the structure.
- Rotational torque from the wind varies periodically within each cycle, and thus unwanted power periodicities appear at the output.
- It normally requires guy ropes attached to the top for support, which could limit its applications particularly for offshore sites.
- It is noisier than HAWT,
- As wind speed increases significantly with height, for the same tower height HAWT captures more power than VAWT the technology is under development stage and far less is known about them ascompared to HAWTs.

HAWT	***VAWT***
Axis of rotation is parallel to the airstream	Axis of rotation is perpendicular to the airstream
These are commonly used and almost fully developed	These are under developed stage.
The rotor has to face wind stream. It is provided with yaw mechanism to keep it facing wind stream	The rotor can accept wind stream from any directions. There is no need of yaw mechanism
Nacelle carrying gear train, controls and generator has to be mounted on top of the tower.	Gear train, controls and generator can be located at ground level.
Tower has to be strong and designed properly	Tower is simple in construction and installation
Inspection and maintenance of wind mill is difficult	Inspection and maintenance of wind mill is easy
Costly	Less costly
Less noisy	More noisy

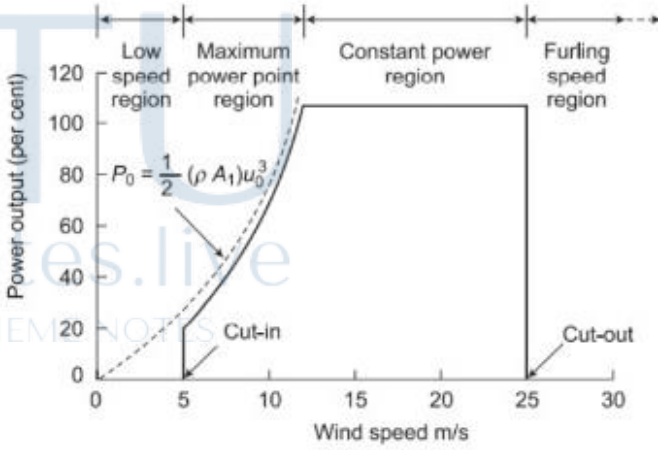
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Extract more power from wind	Extract less power from wind
Technology fully developed	Technology is under development
Less fatigue to parts due to wind action	More fatigue to parts due to wind action
Designed to use lift force	Designed to use drag force
More efficient	Less efficient
Smooth output	Fluctuating output
Produces lower starting torque	Produces high starting torque
Operates properly in moderate wind speeds	Can operates even in low wind speeds
Pitch of blade can be controlled	Pitch of blade cannot be controlled

Wind Turbine Operation and Power versus Wind Speed Characteristics

The power-speed characteristics of a wind turbine have four separate regions as shown in Fig.

1) ***Low Speed Region (Zero to Cut-in Speed)*** In this region, the turbine is kept in braked position till minimum wind speed (about 5 m/s), known as cut-in speed becomes available. Below this speed the operation of the turbine is not efficient.
2) ***Maximum Power Coefficient Region***: In this region, rotor speed is varied with wind speed so as to operate it at constant tip-speed ratio, corresponding to maximum power coefficient, CP max. In this range the nature of characteristics is close to that of maximum power available in the wind. The turbine is operated at maximum power output point using pitch control.



Power versus wind speed characteristics

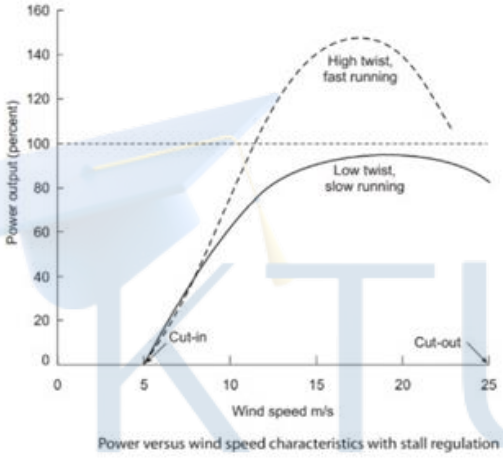
3) ***Constant Power Region (Constant Turbine Speed Region)*** During high speed winds (above 12 m/s), the rotor speed is limited to upper permissible value based on the design limits of system components. In this region the power coefficient is lower than CP max.

Large machines use pitch control to maintain turbine speed as constant. Such a machine is known as **pitch regulated.** Classical European machines use **fixed blades (constant pitch)** and the blade twist and thickness are so designed as to crudely maintain the speed as constant. Such a machine is known as **stall regulated**. This is a simple system requiring only passive technique, as there are

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no moving blade surfaces or complicated hardware. However, power capture is somewhat less due to rounding of power curve. In case of fixed blades rotor, the flow over the blade stalls. The stall performance depends on the twist of blade tip from the root and thickness of blade sections. The lift and drag coefficients change so as to flatten out the peak of the power curve. Thus the power output of the wind turbine is limited as shown in Fig. Stall regulation can be used without much problem up to about 25 m diameter rotor. Above this size severe vibration problems, associated with the stalled flow in high wind speeds, have been encountered. Thus large HAWTs always have variable pitch control. At still higher wind speeds, such as gusts, constant speed operation is maintained by employing additional means such as some kind of braking (e.g. eddy current braking).

4) ***Furling Speed Region (Cut-out Speed and Above)*: Beyond** certain maximum value of wind speed (around 25 m/s) rotor is shut down and power generation is stopped to protect the blades, generator and other components of the system.



Power versus wind speed characteristics with stall regulation

Betz Criterion

In practice **all of the kinetic energy in the wind cannot be converted to shaft power** since the air must be able to flow away from the rotor area. The Betz criterion derived using the principles of conservation of momentum and conservation of energy suggests a maximum possible turbine efficiency, (or power coefficient) of **59 per cent.** In practice power coefficients of 20–30 per cent are more common.

Energy Available in Wind

$$KE = P_0 = \frac{1}{2} \times \dot{m} u_0^2 \tag{6.1}$$

where $\dot{m}$ is the mass of air passing through an area A per unit time and u_0 is the speed of free wind.

The mass of wind can be given as:

$$\dot{m} = \rho A u_0 \tag{6.2}$$

where ρ is the density of air. From Eqs. (6.1) and (6.2), we have

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$$P_0 = \frac{1}{2}(\rho A u_0)\, u_0^2$$

$$= \frac{1}{2}\rho A u_0^3$$

or

$$\frac{P_0}{A} = \frac{1}{2}\rho u_0^3$$

The above relation indicates that the power available in wind per unit area is proportional to the cubic power of its speed.

Terms and Definitions Of Fluid Mechanics

Rotor

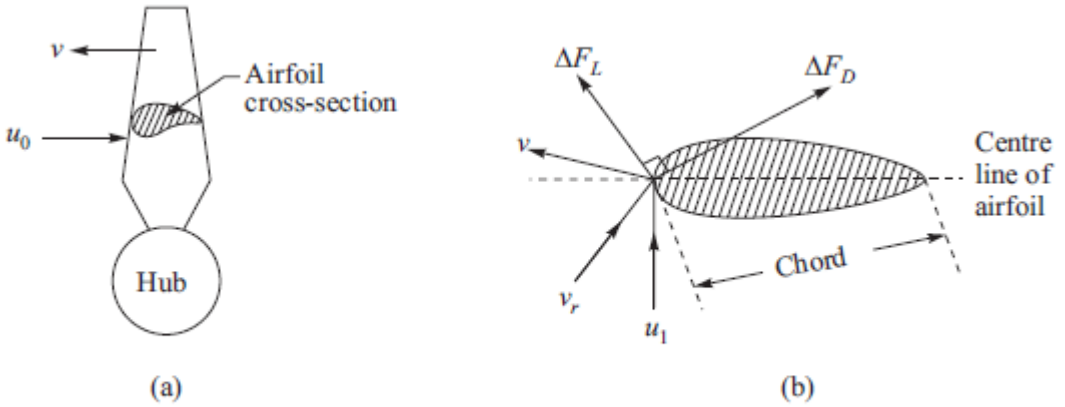
The wind turbine extracts energy from the wind streams by transforming the kinetic energy of the wind into the rotational motion of the rotor of the wind turbine. The wind energy is extracted by the rotor of wind turbine as it has one, two or three blades having aerofoil shape and these blades are attached to hub. The wind moving over the surface of blades (aerofoils) attached to the rotor generates the requisite forces to turn the rotor.

Solidity

It is defined as the ratio of the projected area of the rotor blades on the rotor plane to the swept area of the rotor. High-solidity rotors use drag force for rotation and these rotors turn slower. Low-solidity rotors have slender aerofoil blades and these rotors use lift force for rotation. They turn faster.

Chord

It is the width of the blade which is across distance from one edge of the blade to the other edge as shown in Figure.



Rotor blade and its cross-section. (a) Rotor blade and (b) Aerofoil shape of blade's cross-section.

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Different velocities

Various velocities are as follows:

Wind velocity (u_0): The velocity of free air stream at sufficient distance away from wind turbine where there is no disturbance due to rotation of the wind turbine.

Incident wind velocity (u_1): It is the velocity at which the wind strikes the blade. This velocity is always slightly lower than the free air stream velocity (u_0).

Blade linear velocity (v): It is the tangential velocity of the blade due to the rotation of blade. If ω is the angular velocity of the blade and R is the length of the blade, then blade linear velocity $v = \omega \times R$.

Relative velocity (v_r): It is the relative velocity of wind with respect to the moving blade. It is the vector sum of incident wind velocity (u_1) and blade linear velocity ***(v).***

Mathematically, it can be given as:

$$\vec{v_r} = \vec{u_1} + \vec{v}$$

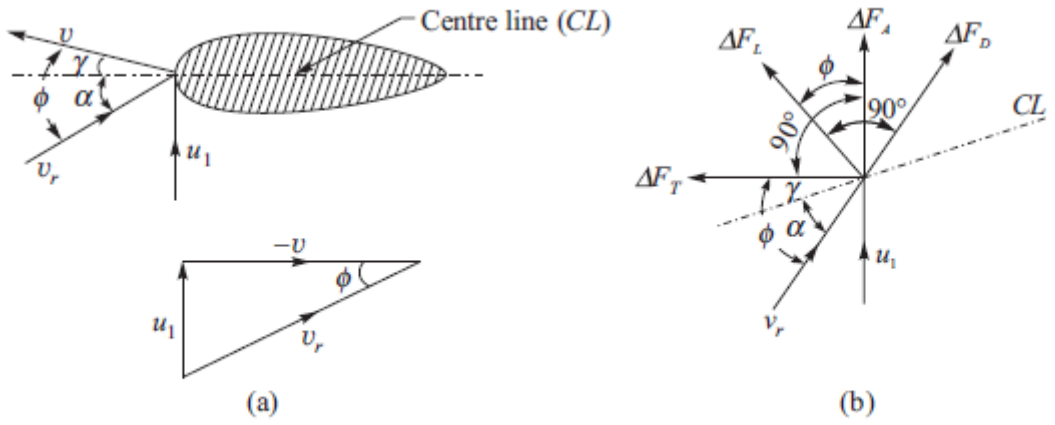
Angle of attack or angle of incidence (α)

It is the angle between the centreline of the aerofoil (blade cross-section and the relative wind velocity vr) as shown in Figure 6.6. The airflow remains attached to the aerofoil for small angle of attack. The airflow is separated from the aerofoil for large angles of attack.

Pitch angle or blade setting angle (γ)

It is the angle between the centreline of the aerofoil and the direction of linear motion of the blade. The output of a turbine is greatly influenced by blade pitch angle. The blade pitch control is a very effective way to control. The output power, speed or torque of the turbine.

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Velocities and forces acting on aerofoil (a) Various velocities and angles of aerofoils, (b) Various forces on aerofoil.

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Drag force (ΔF_D)

When a body (aerofoil) is placed in uniform airflow, there are two forces acting on the body, namely, pressure force acting normally to the surface of the body, and shear force acting along the surface of the body. The pressure force and shear force can be combined to give the total force exerted by the wind on the body. This total force can be now resolved in the direction parallel to airflow and perpendicular to the airflow. The component of total force parallel to the direction of airflow is called drag force. The drag force always opposes the relative motion between the body and the air. It is given by the following equation:

$$\text{Drag} = \Delta F_D = C_D \times \frac{1}{2}\rho u_1^2 \cdot A$$

where C_D is the drag coefficient and A is the projected area of the body aerofoil perpendicular the direction of airflow.

Lift force (ΔF_L)

The component of the total force (pressure force and shear force) on the body in the direction perpendicular to airflow is called lift force. As the name suggests, this force tries to lift the body. It is given by the following equation:

$$\text{Lift} = \Delta F_L = C_L \times \frac{1}{2}\rho u_1^2 \cdot A$$

where C_L is the lift coefficient.

Axial force (ΔF_A)

The total force (pressure force and shear force) can also be resolved along the axis of rotation of blade and perpendicular to it (tangential force on the blade). The component of total force acting on the blade along the axis of rotation of the blade is called the axial force. The axial force does not contribute to the rotation of the blade. It is also called thrust force and has to be balanced by a suitable reaction force generated by any thrust bearing provided on a rotor. The axial force contributes to waste energy which cannot be extracted from wind energy. Hence, axial force should be as less as possible. The axial force can be given as follows:

$$\Delta F_A = \Delta F_L \cos \phi + \Delta F_D \sin \phi$$

where $\phi = \alpha + \gamma$.

Tangential force (ΔF_T)

It is the component of total force on the blade acting tangential to its circular path of rotation. This is the force which contributes mainly to the useful energy extracted from the wind energy. It should be as high as possible. It is given by:

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$$\Delta F_T = \Delta F_L \sin \phi - \Delta F_D \cos \phi$$

Tip speed ratio (λ)

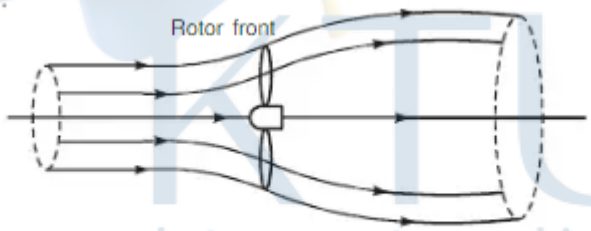
The tip speed ratio is defined as the ratio of the speed of tip of the rotor blade to the speed of oncoming air. Hence, tip speed ratio is:

$$\lambda = \frac{\omega \times R}{u_0}$$

For a particular wind speed, there exists an optimum turbine tip speed to produce the maximum output. It is important to match the rotation of wind turbine to the corresponding wind speed.

WIND ENERGY EXTRACTION

Wind turbines extract energy from wind stream by converting the kinetic energy of the wind to rotational motion required to operate an electric generator. By virtue of the kinetic energy, the velocity of the flowing wind decreases. It is assumed that the mass of air which passes through rotor is only affected and remains separate from the air which does not pass through the rotor. Accordingly, a circular boundary surface is drawn showing the affected air mass and this boundary is extended upstream as well as downstream as detailed in Figure 7.9.



Representation of wind flow through turbine.

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As the free wind (stream) interacts with the turbine rotor, the wind transfers part of its energy into the rotor and the speed of the wind decreases to a minimum leaving a trail of disturbed wind called **wake** [(Figure 7.10(a)]. The variation in velocity is considered to be smooth from far upstream to far downstream. However, the fall in static wind pressure is sharp as depicted in Figure 7.10(b). The wind leaving the rotor is below the atmospheric pressure (in wake region) but at far downstream it regains its value to reach the atmospheric level. The rise in static pressure is at the cost of kinetic energy, consequently further decreasing the wind speed.

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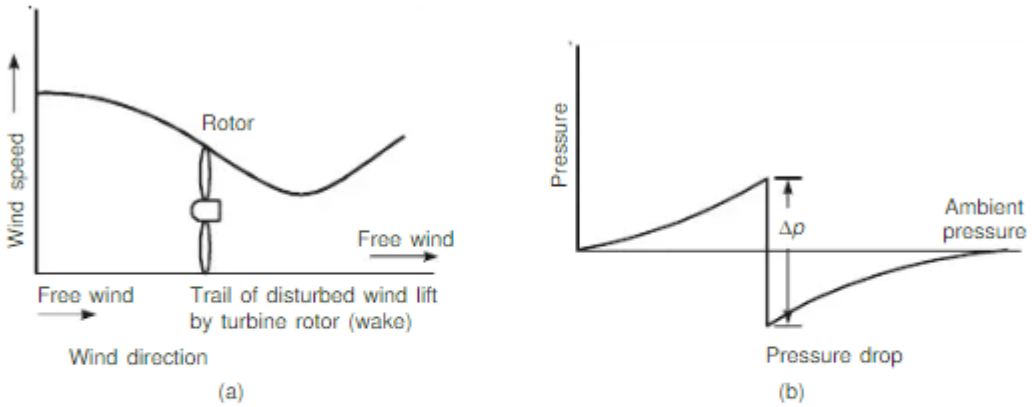


Figure 7.10 Change in wind speed and pressure in traversing the turbine rotor.

Wind flow is considered incompressible and hence the air stream flow diverges as it passes through the turbine. Also the mass flow rate of wind is assumed constant at far upstream, at the rotor and at far downstream. To compute the mathematical relationships, suppose:

P = atmospheric wind pressure
P_u = pressure on upstream of wind turbine
P_d = pressure on downstream of wind turbine
V = atmospheric wind velocity
V_u = velocity of wind upstream of wind turbine
V_b = velocity of wind at blades
V_d = velocity of wind downstream of wind turbine before the wind front reforms and regains the atmospheric level
A = area of blades
$\overline{M}$ = mass flow rate of wind
ρ = air density.

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The kinetic energy of wind stream passing through the turbine rotor is

$$\text{KE} = \frac{1}{2}\overline{M}V_b^2$$

and $$\overline{M} = \rho A V_b$$

Hence, $$\text{KE} = \frac{1}{2}\rho A V_b^3 \quad (7.1)$$

The force on the disc of the rotor can be expressed as

$$F = (P_u - P_d)A \quad (7.2)$$

Force on the rotor can be expressed as change of momentum per unit time from upstream to downstream winds, i.e.,

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$$F = \bar{M}(V_u - V_d) \tag{7.3}$$

Applying the Bernaulli's equation to upstream and downstream sides,

$$P + \frac{1}{2}\rho V_u^2 = P_u + \frac{1}{2}\rho V_b^2 \tag{7.4}$$

$$P_d + \frac{1}{2}\rho V_b^2 = P + \frac{1}{2}\rho V_d^2 \tag{7.5}$$

Solving Eqs. (7.4) and (7.5), we get

$$P_u - P_d = \frac{1}{2}\rho(V_u^2 - V_d^2) \tag{7.6}$$

Equating Eqs. (7.2) and (7.3), we get

$$(P_u - P_d)A = \bar{M}(V_u - V_d) = \rho A V_b (V_u - V_d) \tag{7.7}$$

Solving Eqs. (7.6) and (7.7), we get

$$\frac{1}{2}\rho(V_u^2 - V_d^2) = \rho V_b (V_u - V_d)$$

or

$$V_b = \frac{V_u + V_d}{2} \tag{7.8}$$

In a wind turbine system "Steady Flow Work", W, is equal to the difference in kinetic energy between upstream and downstream of the turbine for unit massflow, $\bar{M} = 1$. Therefore,

$$W = (\text{KE})_u - (\text{KE})_d$$

$$= \frac{1}{2}(V_u^2 - V_d^2) \tag{7.9}$$

The power output P of wind turbine is the rate of work done, using the mass flow rate equation.

$$P = \bar{M}\left(\frac{V_u^2 - V_d^2}{2}\right)$$

$$= \rho A\left(\frac{V_u + V_d}{2}\right)\left(\frac{V_u^2 - V_d^2}{2}\right)$$

$$= \frac{1}{4}\rho A(V_u + V_d)(V_u^2 - V_d^2) \tag{7.10}$$

For maximum turbine output P, differentiate Eq. (7.10) with respect to V_d and equate to zero to obtain

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$$\frac{dP}{dV_d} = 3V_d^2 + 2V_u V_d - V_u^2 = 0$$

The above quadratic equation has two solutions, i.e., $V_d = \frac{1}{3} V_u$ and $V_d = V_u$

For power generation $V_d < V_u$, so we can have only $V_d = \frac{1}{3} V_u$ (7.10a)

Therefore,

$$P_{max} = \frac{8}{27} \rho A V_u^3 \quad (7.11)$$

$$= \frac{16}{27} \left(\frac{1}{2} \rho A V_u^3 \right)$$

$$= 0.593 \left(\frac{1}{2} \rho A V_u^3 \right)$$

Total power in wind stream is

$$P_{total} = \frac{1}{2} \rho A V_u^3 \quad (7.11a)$$

Therefore, $P_{max} = 0.593 P_{total}$

Maximum theoretical efficiency η_{max} (also called the power coefficient C_p) is the ratio of maximum output power to total power available in the wind, i.e.,

$$\text{Power coefficient, } C_p = \frac{P_{max}}{P_{total}} = 0.593 \quad (7.12)$$

The factor 0.593 is known as the **Betz limit** (After the name of the engineer who first derived this relationship).

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Available efficiency

Theoretically, the maximum power extracted by a turbine rotor is 59.3% of the total wind energy in the area swept by the rotor. Considering the rotor efficiency to be 70%, bearing, vibrations, friction losses and generator efficiency 90%, the available efficiency η is 60% of C_p, i.e.,

$$\eta_a = 0.6 \times 0.593$$

$$= 35.5\%$$

Torque Generated by Wind Turbine

The maximum energy of wind is given by P_0 and it exerts torque at the blade of wind turbine. If T_{max} is the maximum torque exerted on turbine rotor at the tip of the blade having radius R, we have

$$T_{max} = F_{max} \times R$$

For a wind turbine which is producing torque T, less than T_{max} we have

$$T = C_T \times T_{max}$$

where C_T is the torque coefficient.

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$$C_T = \frac{C_P}{\lambda}$$

The above relation indicates that both C_T and C_P are functions of λ. The wind turbines havings high tip speed ratios will require low starting torque.

tip speed ratio is

$$\lambda = \frac{\omega \times R}{u_0}$$

Tip Speed Ratio for Maximum Output

A turbine should be designed in such a way that neither of the following conditions is met:

- If the blades are provided in excess or blades are rotated fast, this results in a blade moving in the air turbulence created by a proceeding blade.
- If the blades are provided in less numbers or blades are rotated slowly, this results in a large amount of wind passing through the space between two blades without interacting with the blades.

If a wind turbine has "n" numbers of blades and it is rotating with angular velocity ω, the time (t_1) taken for a blade to move into the previously occupied blade's position is given by

$$t_1 = \frac{2}{n \times \omega}$$

If x is the length of the wind stream's disturbance created by a blade, the time (t_2) taken for this disturbance to move out is given by

$$t_2 = \frac{x}{u_0}$$

The maximum power output by the wind turbine will take place at the blade when we have

$$t_1 = t_2$$

or

$$\frac{2}{n \times \omega} = \frac{x}{u_0}$$

or

$$\frac{\omega}{u_0} = \frac{2}{n \times x}$$

Multiply with R, that is length of blade, we obtain

$$\frac{\omega \times R}{u_0} = \frac{2R}{n \times x}$$

$$\lambda_0 = \frac{2R}{n \times x}$$

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where λ_0 is the tip speed ratio for the maximum output.

In case $x = \frac{R}{2\pi}$, we have

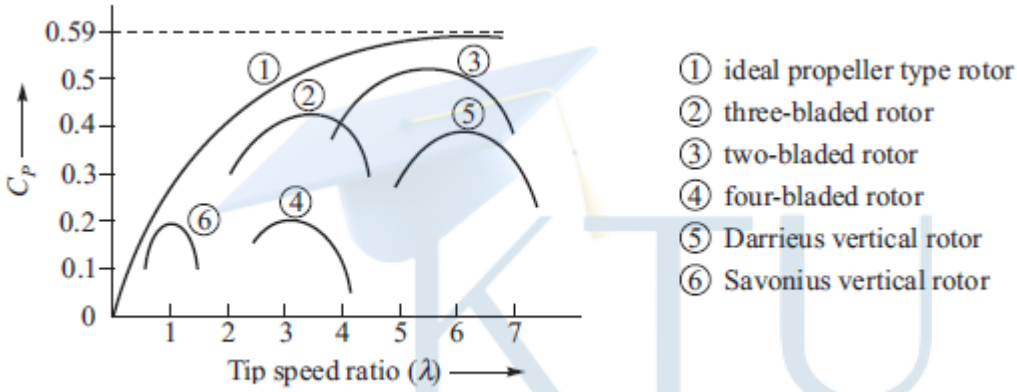
$$\lambda_0 = \frac{4\pi}{n}$$

The maximum power extraction will occur when tip speed ratios are:

(i) $\lambda_0 = \frac{4\pi}{2} = 2\pi$, where $n = 2$

(ii) $\lambda_0 = \frac{4\pi}{4} = \pi$, where $n = 4$

The maximum output is possible when power coefficient $C_P = 0.59$. The relationship between C_P and λ for ideal propeller type rotor and other wind turbines are as shown in the Figure



Variation of power coefficient with tip speed ratio.

Selection of Site for Wind Generator

- ***High annual mean wind speed.***

A basic requirement for a successful use of a windmill or farm is an adequate supply of wind and good wind speeds. The wind power is proportional to the cubic power of wind speed.

- ***No obstruction.***

There should not be any high structure to obstruct wind for a distance of 3 km to the windmilll.

- ***Open plain.***

The site should be the open plain such as open sea shoreline where strong winds prevail.

- ***Height.***

The wind speed increases with height, which can be obtained when the windmill is located on a hill or a ridge with gentle slope.

- ***Near lake or ocean.***

Differential heating of water and land generates wind of sufficient speeds.

- ***Topography.***

Topography such as mountain gap helps to channelise and speed up winds.

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- ***Favourable land cost***

It helps in restriction or reducing the cost of project.

Nearness to load centre.

It reduces the cost of transmission of the generated power.

Nearness to road or rail link.

It helps in installation of windmill.

Availability of wind rose.

It helps in designing of windmill as wind data of the site can be determined.

University Questions

1. List the factors that influence the site selection for wind power plant.
2. Briefly explain the concept of power coefficient and Betz limit with reference to a wind turbine
3. With neat diagrams, differentiate between Horizontal Axis Wind Turbine and Vertical Axis Wind Turbine. Also, compare their relative advantages and disadvantages.
4. Explain the general block diagram of a grid connected Wind Energy Conversion system
5. Using relevant diagram, explain the terms pitch angle, angle of attack, solidity, relative velocity, lift force and drag force in the context of wind energy conversion systems.
6. Explain the output power versus wind speed characteristics of a wind turbine.
7. Define tip speed ratio, cut in speed and cut out speed of a wind turbine
8. Explain the term capacity factor for a wind power plant.
9. With the help of a diagram, explain the construction and working of Horizontal Axis Wind turbine.
10. Explain the term "Betz Criterion" and derive an expression for Power Coefficient. Obtain its maximum value
11. With the help of a diagram, explain the construction and working of Vertical Axis wind turbine.
12. Differentiate between HAWT and VAWT.
13. Explain with illustration the power output versus wind speed characteristics of a wind turbine
14. Derive the expression for maximum power extracted by a wind turbine.
15. Discuss the importance of Kyoto protocol in sustainable development.
16. Discuss the relationship between energy and sustainable development.
17. Explain the need to increase the utilization of non-conventional energy sources
18. Write notes on world energy scenario
19. Compare the advantages of non-conventional sources of energy with conventional sources.
20. List the non-conventional energy resources and explain the merits and demerits
21. Explain the concept of sustainable development with reference to environment, economy and Energy.
22. List the various non-conventional energy resources
23. Explain the advantages and limitation of each non-conventional energy resource
24. Explain how the promotion of renewable energy sources helps in attaining sustainable development goals.

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25. Compare the availability and limitations of conventional and non-conventional energy resources in the context of sustainable development. How can the increased use of non-conventional energy resources mitigate environmental impacts while supporting economic growth?
26. List the disadvantages of conventional energy sources.
27. Write down the disadvantages of solar power and wind power generation.
28. Write notes on the current Indian and World Energy Scenario.

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